Jiale Shao

Vehicle Engineering | CAD Design | Simulation & Validation

London, UK | jiale.shao24@imperial.ac.uk | linkedin.com/in/jiale-shao

MEng Design Engineering student at Imperial College London with aerodynamic package development experience at **Chery Racing Development**, and CAD, CFD and FEA project work backed by physical testing. Keen to bring design, simulation-validation and build follow-up skills to a vehicle engineering team.

EDUCATION

Imperial College London — MEng Design Engineering 2024 – 2028 (Expected)

Modules: Fluid & Thermal Dynamics, Solid Mechanics, Finite Element Analysis, Electronics & Circuit Design, Robotics, Computer Science, Human-Centred Design Engineering

Awards: **WRO (World Robot Olympiad) Provincial Champion(2023)**; **School Principle's First-Class Scholarship** (Top 3% in high school from 2021-2024)

PROFESSIONAL EXPERIENCE

Chery Racing Development, Chery Automobile — Aerodynamics Intern Jul – Sep 2026

Global Technology Innovation Center

- **Set** downforce and drag targets for a 330 electric performance sedan (Exlantix ES-based), targeting Nürburgring lap attempts, and **designed in CAD** the front splitter, side skirts, swan-neck rear-wing mounts and adjustable wing-angle mechanism for its next-generation aero package.
- **Worked closely with a CFD teammate** in the Aerodynamics department on iterative STAR-CCM+ optimisation (**CL −0.155 → −0.565**); established a front/rear **CLF/CLR distribution methodology**, benchmarked against Porsche Taycan Turbo GT with Manthey Kit Nürburgring data.
- **Followed** first-generation (V1) carbon aero-kit **manufacturing progress at the factory** with a colleague, linking design intent to what was being built.
- **Led** FIA/WSC technical-regulation analysis for an early-stage TCR project, classifying the splitter, rear wing and diffuser design space into **three levels of freedom** (fixed / approval-required / free) and drafting the CFD methodology (project paused for resourcing before simulation); researched Formula E Gen4, WRC Rally1 and Le Mans Hypercar regulations.

ENGINEERING PROJECTS

Electric GT Car Aerodynamic Optimisation — Personal Project

- **Built** a 1:17 Ansys Fluent CFD model; refined the nose, underbody Venturi tunnel and Kamm-tail, cutting simulated **CD by 35%** (0.71 → 0.50).
- **Validated** with a 1:21 3D-printed model on an aerodynamic balance in a wind tunnel (surface tufting + force measurement): **CD 0.34, CL −0.17**; analysed the CFD–test gap through Reynolds-number effects.

Prosthetic Blade Structural Optimisation — Personal Project

- **Raised** the first natural frequency from **16 Hz to 51.7 Hz** in Ansys Workbench to avoid running resonance; **validated** with static FEA (**safety factor 16.6**) and **Gerber mean-stress** fatigue assessment.

Hachi-Pot Bionic Smart Planter — Team Project

- **Designed** the layered enclosure and servo-driven expression mechanism in Fusion 360 for an ESP32-based interactive planter, dividing CAD, electronics and coding across the team around members' availability.

Shell Eco-marathon (ICE Powertrain) & RC Racing Car — Imperial College London / Team Projects

- **Reverse-engineered** an internal-combustion engine with teammates through teardown, measurement and Fusion 360 reconstruction; on an RC racing car, **led structural design and modelling iterations** while teammates handled programming.

TECHNICAL SKILLS

CAD / CAE: SolidWorks, Fusion 360, Ansys Fluent, Ansys Mechanical/Workbench, STAR-CCM+ (via team collaboration)

Engineering: Vehicle aerodynamics, FEA, design validation, reverse engineering, DFM, technical-regulation analysis

Programming & Prototyping: Python, MATLAB, C++; FDM/SLA 3D printing, ESP32, Arduino

Languages: Chinese (native), English (advanced)